

Bluetooth Low Energy Based Automatic Attendance System

Abstract—This paper presents a secure Bluetooth Low Energy (BLE)-based attendance system aimed at preventing proxy attendance and improving efficiency in academic settings. Professors send BLE packets through a dedicated Android app, which students scan to mark their attendance through a MongoDB-connected back-end. The system ensures authenticity using Firebase and biometric log-in, Android ID binding, and SIM subscription ID checks. BLE's limited range enforces physical presence, while the backend verifies enrollment and prevents duplicate entries. This approach enhances trust and offers scalable deployment.

I. INTRODUCTION

Traditional attendance systems in educational institutions are often vulnerable to inefficiencies, manipulation, and proxy attendance, compromising both academic integrity and administrative efficiency. This proposal introduces a secure, scalable and technology-driven solution: a Bluetooth Low Energy (BLE)-based attendance system that leverages proximity-based validation, biometric authentication, device-level binding along with Sim binding to ensure authentic presence recording.

At the core of this system is a multi-role Android application designed for professors, students, and administrators. Professors initiate attendance by broadcasting a BLE signal embedded with a secure MongoDB record ID specific to a session. Students within range receive the broadcast and respond via the app, which then communicates with the backend server to validate and mark attendance. The backend verifies multiple parameters, such as student-course enrollment, duplication attempts, and session status, before updating the record.

To mitigate proxy attendance, the system integrates biometric login, Android device ID binding, and SIM subscription ID binding, ensuring that only authorized individuals using preregistered devices can interact with the platform. The inherent limitation of the physical range of BLE adds an additional layer of security by requiring the physical presence of the student within the classroom.

This solution offers a significant advancement over Biometric Hardware or manual attendance systems by being more secure, contactless, real-time, and scalable. Designed for educational institutions, but adaptable to corporate, this system addresses a clear gap in secure attendance verification, offering strong potential for institutional deployment and commercial licensing.

II. PROBLEM STATEMENT

Attendance tracking is a fundamental aspect of academic administration, but ensuring its accuracy and integrity remains a persistent challenge. In most educational institutions, existing attendance methods are either outdated or prone to manipulation. Proxy attendance, where students mark attendance on behalf of others—is a widespread issue that undermines both academic accountability and institutional credibility. This problem becomes more pronounced in larger classrooms, making manual supervision impractical. Traditional systems like manual roll calls, RFID-based card swiping, and Biometric fingerprint are commonly used to manage attendance.

A. Manual Roll Call

Manual roll call requires instructors to call out student names and mark attendance based on verbal responses. While simple and cost-effective, it is time-consuming and prone to proxy attendance, as students can easily respond for absent peers. This method lacks any form of automation or verification.

B. RFID Based Cards

RFID systems automate attendance by allowing students to scan ID cards. However, they are vulnerable to misuse since cards can be shared among students. The need for dedicated hardware and ongoing maintenance adds to institutional overhead, limiting scalability.

C. Biometric Hardware

Biometric fingerprint scanners provide identity verification but require expensive hardware and centralized infrastructure. These systems cause delays, raise hygiene concerns, and are impractical for flexible or off-site classroom settings. Moreover, they cannot confirm a student's presence in the correct location or at the correct time.

The central issue across all these methods is the absence of context-aware, presence-verified, and identity-bound attendance. Most existing systems operate under the assumption that students interact honestly with the technology, but this is rarely enforced technically. The lack of secure identity binding, proximity validation, and device-level controls opens the door to repeated misuse, eroding trust in academic records and complicating data auditing.

III. PROPOSED SYSTEM

The proposed attendance system is a secure, proximity-aware application that leverages Bluetooth Low Energy (BLE), biometric validation, and cloud-based identity verification to eliminate proxy attendance while maintaining high usability and backend scalability. The system is developed using Jetpack Compose (Kotlin) for Android, Node.js with Express for backend logic, MongoDB for data persistence, and Firebase for authentication. The backend is deployed in a serverless AWS environment for scalability and reliability.

A. System Roles and Authentication Flow

The mobile application operates in three modes: *Student*, *Professor*, and *Admin*. Upon launch, the application requests BLE and location permissions essential for proximity-based interactions. There is no user-facing registration; account creation is solely performed by the Admin through manual entry or CSV upload. Firebase handles login authentication and issues a unique user identifier (UID). This UID is verified by the backend to determine user role and validate access.

For student login, additional verification is performed using the Android device ID and SIM subscription ID. The backend ensures that each UID is mapped to a single device and SIM. If mismatched, the login is rejected. In cases where a new device is used, a new mapping is created only if the account is authorized. On successful device verification, biometric authentication is triggered to complete the login.

B. Professor Workflow and BLE Broadcasting

Professors can create courses by specifying metadata such as batch, academic year, and expiry date. A unique six-character course joining code is generated using `nanoid` [5], acting as a secure identifier for enrollment. After login, professors can initiate BLE broadcasting for a selected course. The app checks for Bluetooth, location, and internet availability before sending a request to the backend to create a new attendance session. The backend responds with a unique record ID, which is broadcasted over BLE using a predefined UUID, isolating it from unrelated nearby BLE signals.

Broadcasting lasts for 45 seconds by default, during which nearby students can scan and mark attendance. Professors can also manually stop or resume broadcasting. After the session, they can view students who missed marking and update attendance manually if needed. Attendance records can be exported in Excel format.

Professors can also access archived courses—automatically transferred to a separate collection post-expiry using a scheduled `node-cron` [6] job. While new attendance cannot be marked for archived courses, professors can review attendance logs, student lists, and download reports for audits or records.

BLE broadcasting is implemented using Android's `BluetoothAdapter` [1] and `BluetoothLeAdvertiser` [2] classes via the system `BluetoothManager` [4] service.

C. Student Workflow and BLE Scanning

Students view only active courses they have joined via valid joining codes. Archived courses are hidden to maintain a clutter-free interface.

To mark attendance, students enable Bluetooth and location services, then press a "Scan Attendance" button. The app scans for 10 seconds using Android's `BluetoothLeScanner` [3], filtering packets by UUID. Upon detecting a valid packet (UUID + record ID), the data is sent to the backend, which verifies course enrollment, checks for duplicates, and updates the attendance model. The updated status is shown in the UI instantly.

BLE scanning on the student side is handled via `BluetoothLeScanner` [3], managed through the `BluetoothManager` [4] and `ScanCallback` API.

D. Admin Operations

The Admin interface facilitates centralized control over user and course management, ensuring system integrity and operational consistency. Admins possess the exclusive ability to create and manage student and professor accounts—either through bulk uploads using CSV files or via structured manual entry forms. This approach eliminates the risk of unauthorized or fraudulent account creation, which can compromise the authenticity of the attendance records.

Allowing only admins to onboard users ensures that every account is verified against institutional records, maintaining a trusted environment. By restricting self-registration, the system mitigates the threat of impersonation, multiple fake accounts, or unintended role misuse.

Upon account creation, passwords are auto-generated using a deterministic method—specifically, extracting the string portion before the '@' symbol from the user's email. For security, both students and professors are prompted to change this default password upon their first login to enforce secure credential management practices.

Additionally, admins can:

- View and manage all active and archived courses.
- Inspect real-time or historical attendance records for any course.
- Download attendance logs in Excel format for reporting or auditing.
- Monitor and audit individual student attendance across multiple courses.

Admin accounts are hardcoded into the application and are not part of the public login or registration flow, ensuring they remain inaccessible to unauthorized users.

E. Security and Integrity Measures

Security is implemented across multiple layers:

- Firebase Authentication for secure credential management.
- Android ID and SIM subscription binding for device-level control.
- Biometric authentication for identity verification.

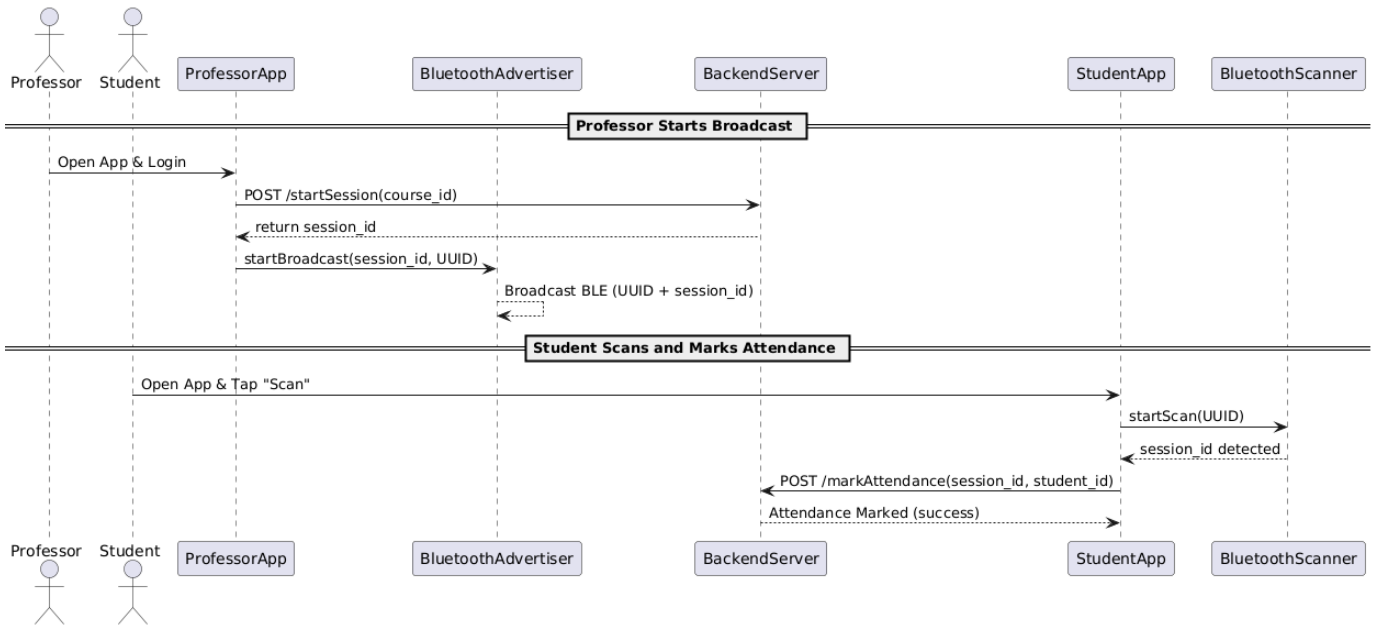


Fig. 1. Sequence diagram showing BLE-based attendance flow between student, professor, and backend.

- BLE UUID-based packet filtering to prevent spoofing.
- Backend-side duplication checks and enrollment validation.

These measures prevent impersonation, device reuse, and packet-level spoofing.

F. Usability and Interface Features

The application is designed for simplicity and clarity, allowing users to operate it intuitively without prior instructions. On launch, users are guided through permission requests and role selection (Student, Professor, Admin) with minimal input required.

Built using Jetpack Compose, the UI is responsive and consistent, ensuring fast interactions and smooth transitions. Biometric authentication integrates seamlessly into the login process, adding security without additional steps.

Key features include:

- Real-time attendance updates visible to students and professors.
- Automated archiving of expired courses for better organization.
- Sort and search options across all lists (e.g., attendance, courses, users).
- Clean, minimal screens with clearly labeled buttons and intuitive navigation.

The system prioritizes ease of use and quick adoption across all roles, requiring no external training or manuals.

G. Robustness and Scalability

Data separation between active and archived courses ensures high performance for real-time operations. The use of serverless backend deployment (AWS) and NoSQL data models enhances scalability. Cron jobs ensure timely cleanup

of expired records, and the application structure supports dynamic classroom environments without dependency on fixed infrastructure like RFID readers or biometric stations.

H. Limitations and Trade-offs

While the system introduces significant improvements in automation, accuracy, and ease of attendance tracking, a few limitations persist due to the inherent constraints of BLE technology and Android's biometric API design.

1. BLE Range and Physical Proximity: A student standing just outside the classroom may still be able to mark attendance if within the BLE broadcast radius. However, broadcasting is limited to 45 seconds and can be triggered at unpredictable times, reducing the chances of remote attendance. Moreover, any sudden crowding or rush near the classroom is easily observable by the professor, serving as a visual deterrent. This setup enforces proximity to the class and discourages habitual bunking.

2. Biometric Verification Limitations: Android's biometric system does not identify which fingerprint was used for authentication—it simply returns a success or failure response. This introduces the theoretical risk of proxy attendance if a student stores a friend's fingerprint on their phone.

However, this comes at a significant personal cost: storing someone else's fingerprint compromises the phone owner's own device security and privacy. Most users are unlikely to risk full access to their phones for minimal academic gain.

Time Efficiency: One of the most significant advantages of the system is the substantial time saved during attendance collection. Traditional methods—manual roll calls, paper sheets, or signing attendance registers—often take several minutes per session, especially in large classrooms. With BLE broadcasting and biometric verification, the entire process completes

within seconds, without disrupting the flow of lectures. This not only maximizes instructional time but also reduces the administrative burden on professors and eliminates the need for post-lecture attendance reconciliation.

IV. CONCLUSION

This paper presents a secure and efficient BLE-based attendance system leveraging biometric verification and short-range broadcasting. By minimizing manual overhead and enhancing accuracy, the system streamlines attendance tracking for both students and faculty. Despite certain technical limitations inherent to BLE and Android biometric APIs, the overall design significantly reduces common loopholes seen in traditional methods. The user-friendly interface, automated archiving, and admin-level control ensure reliability, scalability, and ease of integration within academic institutions.

Overall, despite these edge cases, the system still effectively eliminates major traditional loopholes such as QR sharing, OTP forwarding, and manual signature fraud. The use of short BLE windows and biometric validation provides a strong balance between security and usability.

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